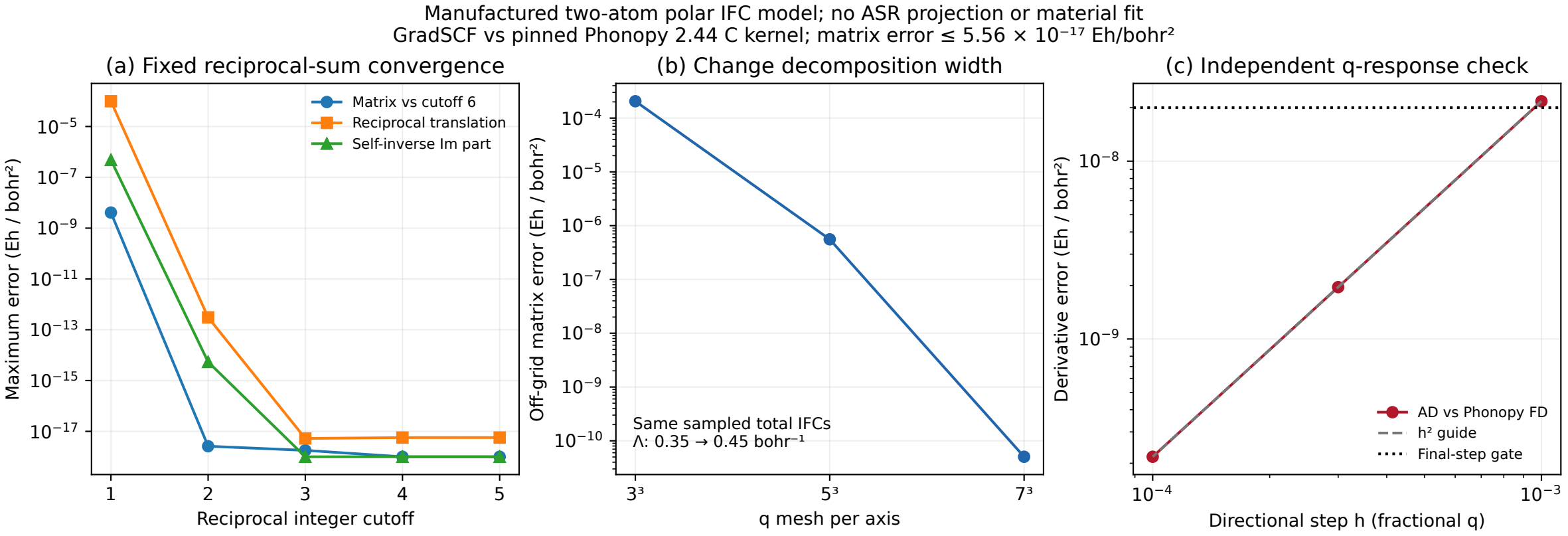
Manufactured two-atom polar IFC model; no ASR projection or material fit
GradSCF vs pinned Phonopy 2.44 C kernel; matrix error ≤ 5.56 × 10⁻¹⁷ Eh/bohr²
(a) Fixed reciprocal-sum convergence
Matrix vs cutoff 6
Reciprocal translation
Self-inverse Im part
Maximum error (Eh / bohr²)
Reciprocal integer cutoff
(b) Change decomposition width
Off-grid matrix error (Eh / bohr²)
Same sampled total IFCs
Λ: 0.35 → 0.45 bohr⁻¹
q mesh per axis
(c) Independent q-response check
Derivative error (Eh / bohr²)
AD vs Phonopy FD
h² guide
Final-step gate
Directional step h (fractional q)